Rome Climate Monitor

Deployment Guide for Total Beginners

No IT knowledge required

[redacted]	Time [redacted]	45-60 min [redacted] Cost	Free [redacted] Skills	None Result	Live webs

What you will deploy

Your project has two parts that work together:

Frontend Map (already live)

The map people see. On Netlify already.

`vocal-marshmallow-7db69a.netlify.app`

Backend API (deploy today)

Fetches satellite data, computes temperatures.

`your-api.up.railway.app`

Services you will use (all free)

Service	What it does	Website	Cost
GitHub	Stores your code	`github.com`	Always free
Railway	Runs your backend	`railway.app`	Free 500h/mo
Copernicus	Real satellite images	`dataspace.copernicus.eu`	Always free
Netlify	Hosts your map	`netlify.com`	Already set up

PART 1 — Create Your Accounts (15 min)

Create these 3 free accounts first. You only need to do this once.

1 Create a GitHub account (stores your code)

- Go to github.com and click the green "Sign up" button
- Enter your email, choose a password and a username
- Verify your email address (check your inbox)
- When asked about team size, choose "Just me"

> GitHub is like Google Drive but for code. Railway needs to read your code from here.

2 Create a Railway account (runs your backend)

- Go to railway.app
- Click "Login" → choose "Login with GitHub"
- Click "Authorize railway-app" when asked
- Done — Railway and GitHub are now linked

3 Create a Copernicus account (real satellite data)

- Go to dataspace.copernicus.eu
- Click "Register" in the top right corner
- Fill in your name and email, create a password
- Check your inbox and click the confirmation link

> This is the European Space Agency free service for Sentinel-2 satellite images of Rome.

PART 2 — Get Your Satellite API Keys (10 min)

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Generate your Copernicus API keys

- Go to dataspace.copernicus.eu and log in
- Click your name/avatar in the top right → "User settings"
- Find "OAuth clients" or "API Access" section
- Click "+ Add client" → name it "rome-climate-api" → type "Confidential"
- Click Save — two codes appear: Client ID and Client Secret
- COPY BOTH and save them in a text file RIGHT NOW

> WARNING: The Client Secret is shown ONLY ONCE and cannot be recovered. Copy it immediately before closing the page.

PART 3 — Upload Code to GitHub (10 min)

5 **Install GitHub Desktop (free app, no command line needed)**

- Go to desktop.github.com
- Click "Download for Mac" or "Download for Windows"
- Install it and sign in with your GitHub account

6 **Upload your project**

- In GitHub Desktop click File → Add local repository
- Choose your rome-climate-backend folder
- If it says "not a Git repository" → click "create a repository"
- In the Summary box type: First commit
- Click the blue "Commit to main" button
- Click "Publish repository" — make sure "Keep private" is UNCHECKED
- Click "Publish Repository"

> Verify: go to github.com/YOUR_USERNAME/rome-climate-backend — you should see your files.

PART 4 — Deploy on Railway (10 min)

7 Create your Railway project

- Go to railway.app and log in
- Click "New Project" (purple button, top right)
- Select "Deploy from GitHub repo"
- Choose rome-climate-backend from the list
- Railway starts building automatically — this takes 5-8 minutes

> The first build takes 5-8 min (installing satellite tools). Go make a coffee!

- When you see a green checkmark — done

8 Add Redis (memory system)

- In your Railway project click "+ New"
- Select "Database" → "Add Redis"
- Railway sets it up automatically in ~30 seconds
- No configuration needed — it connects automatically

PART 5 — Add Secret Keys to Railway (5 min)

Go to Railway → your API service → "Variables" tab. Add each row below:

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Enter all environment variables

- Click the Variables tab in your Railway API service (NOT the Redis one)
- Click "+ New Variable" for each row in the table:

Variable	Value to enter
`APP_ENV`	production
`LOG_LEVEL`	INFO
`COPERNICUS_CLIENT_ID`	paste your Client ID from Part 2
`COPERNICUS_CLIENT_SECRET`	paste your Client Secret from Part 2
`ALLOWED_ORIGINS`	https://vocal-marshmallow-7db69a.netlify.app
`MAX_CLOUD_COVER_PCT`	30
`CACHE_TTL_WEATHER`	1800
`CACHE_TTL_SATELLITE`	43200
`CACHE_TTL_DISTRICTS`	3600
`TILE_SIZE_PX`	512
`ROME_BBOX_LON_MIN`	12.35
`ROME_BBOX_LAT_MIN`	41.78
`ROME_BBOX_LON_MAX`	12.62
`ROME_BBOX_LAT_MAX`	41.98

REDIS_URL is set automatically by Railway — do NOT add it manually.

PART 6 — Get Your Backend URL (2 min)

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Find and verify your live backend URL

- In Railway → click your API service → "Settings" tab
- Look for "Domains" → click "Generate Domain" if nothing is there
- You will see a URL like: rome-climate-backend-production-xxxx.up.railway.app
- Copy this URL — you need it in Part 7!

> Verify it works: open a browser tab and go to: https://YOUR_URL.up.railway.app/api/v1/health
> You should see: {"status": "ok", ...}

PART 7 — Connect the Map to the Backend (5 min)

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Edit one line in the map HTML file

- Find rome-climate-map.html in the frontend/ folder
- Open it with TextEdit (Mac) or Notepad (Windows)
- Press Ctrl+F (or Cmd+F) and search for: const API_BASE = '';
- Change that line to: const API_BASE = 'https://YOUR_RAILWAY_URL.up.railway.app/api/v1';
- Save the file

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Upload updated map to Netlify

- Go to app.netlify.com and log in
- Click your site (vocal-marshmallow-7db69a)
- Click the "Deploys" tab
- Drag and drop your updated rome-climate-map.html file onto the deploy area
- Wait ~30 seconds

> Your map is now connected to the real backend! Open it and check the Legend card at the bottom left.

Troubleshooting

■ Error on the map

Backend is still starting up. Wait 2 minutes and refresh.

■ Map shows fake data

Check that you saved the API_BASE line correctly in the HTML file.

■ Railway build failed

Railway → service → Deployments → click failed deploy → read the red error.

■ Copernicus credential error

Double-check CLIENT_ID and CLIENT_SECRET in Railway variables. No spaces.

■ Lost Client Secret

Copernicus → Profile → OAuth clients → delete old one → create new → update Railway.

■ Yellow badge instead of green

Normal! No clear-sky satellite pass recently. Data is still real weather data.

Congratulations! Your Rome Climate Monitor is now a live website powered by real ESA satellite data. It updates itself every day automatically.